

CSCE 670 · DATA MINING & ANALYSIS · FINAL PROJECT

Instacart Market Basket Analysis: Sequential & Associative Pattern Mining

Combining FP-Growth and PrefixSpan to uncover the hidden temporal rhythm of grocery shopping behavior across 3.4 million orders.

Khussal Pradhan

UID: 437005859

Texas A&M University · Spring 2026

The Grocery Blind Spot

Why today's recommendation engines are fundamentally incomplete

 **Single-Basket Thinking:** Current systems only analyze what's in today's cart.

FP-Growth finds co-purchases but ignores the order of purchases over time.

 **No Temporal Memory:** They ignore the purchasing timeline of individual users.

Flour on Monday → eggs next week — no current system captures this.

The Cost: Wasted perishable inventory and millions in missed revenue.

Our EDA confirmed >99.9% sparsity — most items in <1% of baskets.

3.4M

ORDERS

200K+

USERS

50K+

PRODUCTS



Predicting the Future Cart

Two complementary mining approaches on real Instacart data

-  **Real-World Scale:** 778K+ transactions across 5,000 sampled users.
Reproducible sampling (seed=42) from the 3.4M order Instacart dataset.
-  **Two-Pronged Approach:** FP-Growth for co-occurrence + PrefixSpan for sequences.
Course technique: "bought together" · External: "bought next".
-  **Basket Trajectories:** Chronological sequence of canonicalized basket tuples.
Proper nested structure preserving order boundaries.

WEEK 1
🍌 Banana, 🥛 Milk



WEEK 2
🍌 Banana, 🥚 Eggs



WEEK 3
🍌 Banana, 🍓 Berries

The 7 & 14-Day Rhythm

EDA on 200K+ users reveals strict, predictable shopping cycles



Distinct reorder spikes at **7, 14, 21, and 30 day** intervals — strict weekly rhythms.



Over **99.9% sparsity** — most of 50K products appear in <1% of baskets.



Early-week vs late-week shoppers show **distinct** basket compositions.

What We Found & Where It Goes

Concrete results from 778K+ real transactions

344 **Association Rules** via FP-Growth at 0.5% support across 56K baskets

10K+ **Sequential Patterns** mined by PrefixSpan across basket trajectories

133 **Temporal-only pairs** — bought in sequence but never in the same basket

3 **Research Questions** fully answered: thresholds, segments, sequences

 **The Takeaway:** PrefixSpan found 133 item pairs FP-Growth is blind to — products that follow each other across orders but never appear together.

Example: Organic Raspberries → Organic Zucchini — never co-purchased, but consistently bought in sequence.

 **The Opportunity:** Pre-stock warehouses using 7-day cycle forecasting, push perfectly timed coupons, build time-aware recommendations.

Transform millions of grocery receipts into a roadmap for predictive behavioral logistics.